



Universidad del Desarrollo

Class 7: Feature Engineering

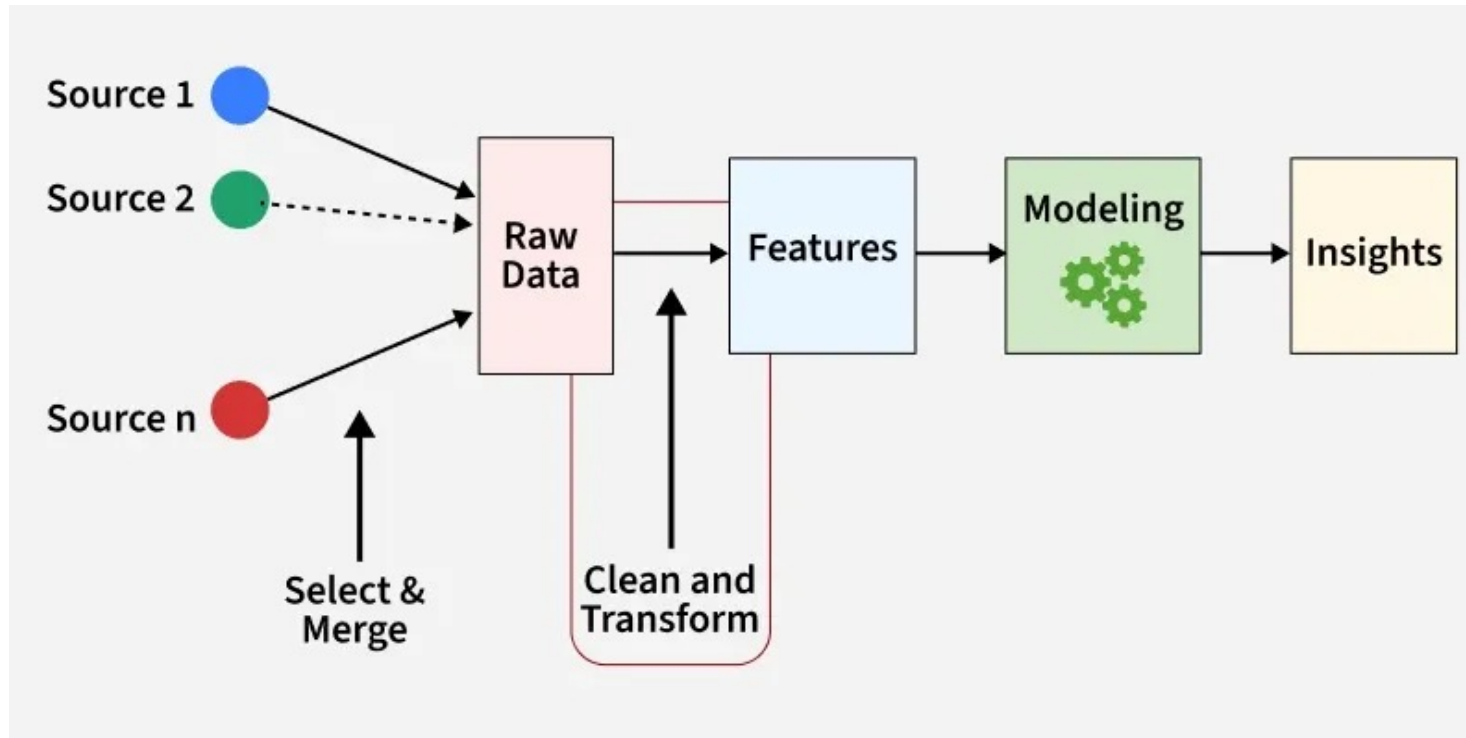
Unit II: Data Preparation and Predictive Models

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Ingeniería

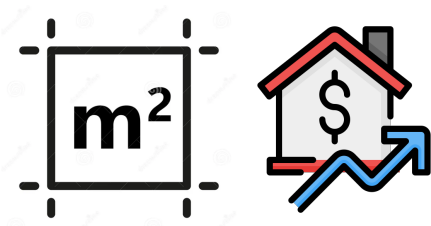

Feature Engineering

Feature engineering is the art and science of transforming raw data into meaningful inputs that improve model performance.

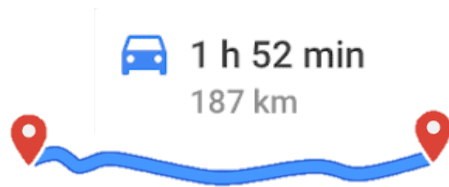


Creating new features

Derive new variables from existing data using mathematical operations, combinations, or decompositions. Computing a price-per-square-meter ratio from price and area, extracting the hour from a timestamp, or calculating the age of an account in days.



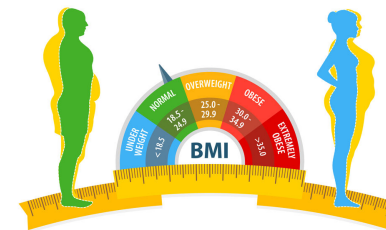
Computing Price per m2



Computing the distance/time between two geographic points using latitude and longitude



Extracting the hour of a time-stamp



Calculating the body mass index (BMI) by combining height and weight

Encoding categorical variables

One-Hot Encoding

Converts categories into binary columns.

Label Encoding

Assigns a unique integer to each category.

Ordinal Encoding

Maps categories to ordered integers.

CATEGORICAL DATA ENCODING TECHNIQUES

Binary Encoding

Converts categories into binary digits.

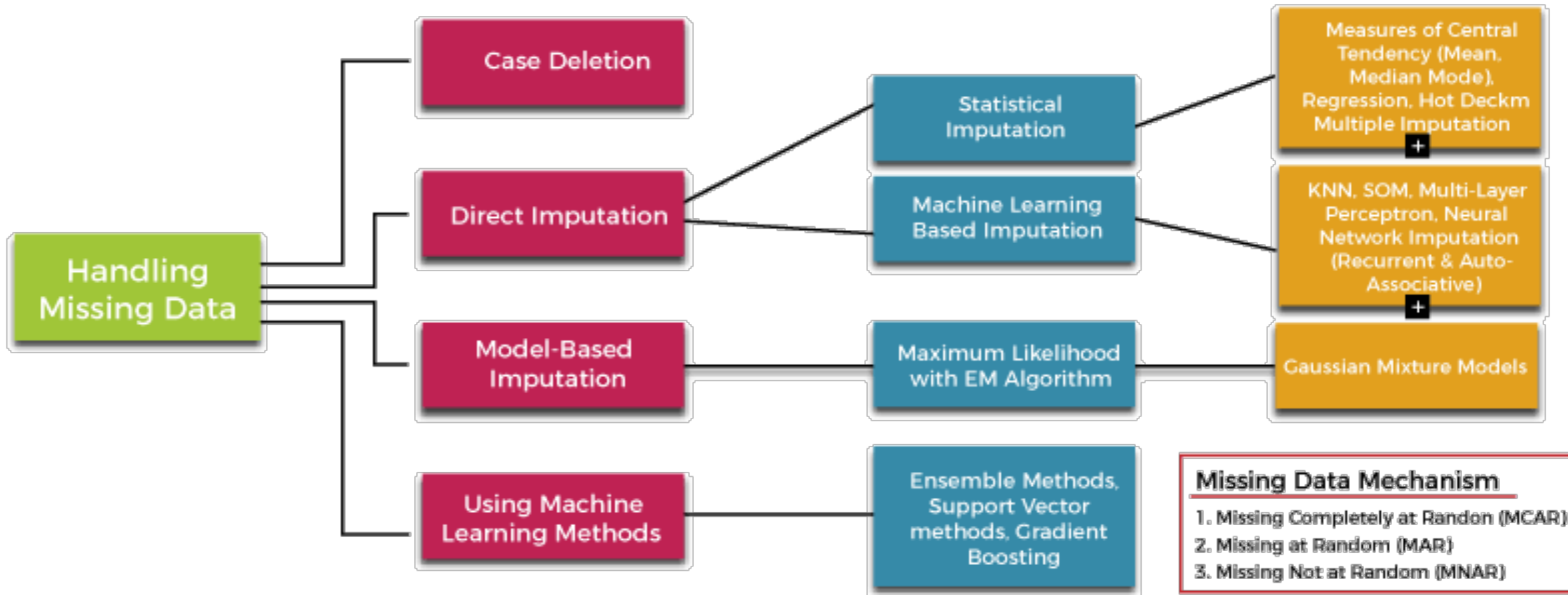
Count (Frequency) Encoding

Replaces categories with their frequency or count.

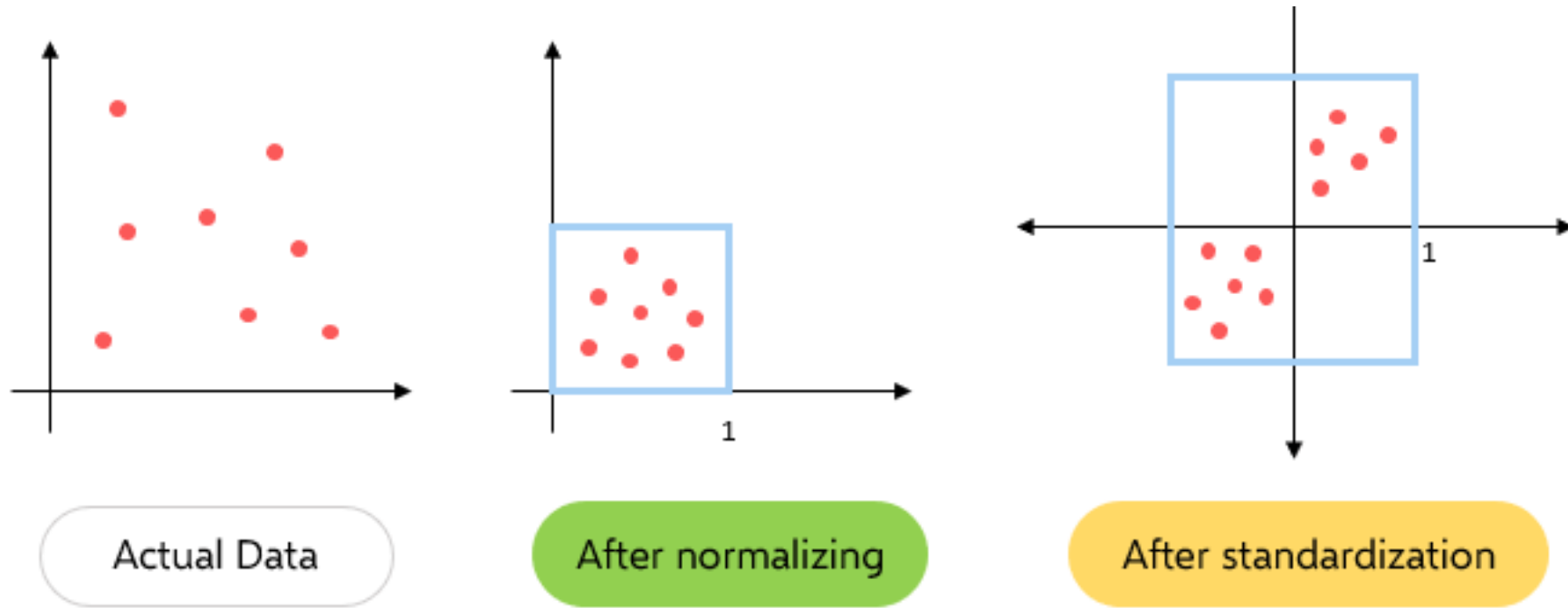
Target Encoding

Replaces categories with the mean target value for each category.

Handling Missing Data ✓



Feature Scaling and Normalization ✓



Bring features to comparable ranges so no single variable dominates. Min-max scaling maps values to a fixed range like 0 to 1. Standardization centers data around zero with unit variance. Robust scaling uses median and interquartile range to handle outliers.

Feature Selection

Identify the most relevant features and discard the rest to reduce noise and overfitting.

Filter-based Approach



Information Gain

Information Gain is a statistic that measures the reduction in entropy (uncertainty) for a specific feature by dividing the data according to that characteristic.



Chi-square Test

The Chi-square test is a statistical test that is used to assess the relationship between two categorical variables.



Fisher's Score

Fisher's Score is a feature selection approach that ranks features based on their ability to differentiate various classes in a dataset.



Missing Value Ratio

The Missing Value Ratio is a straightforward feature selection method that makes decisions based on the number of missing values in a feature.

Wrapper-based Approach



Forward Selection

In forward selection, you start with an empty feature set and iteratively add features to the set.



Backward Selection

The opposite of forward selection is backward selection. You begin with the entire feature set and gradually eliminate features from it.



Exhaustive Feature Selection

Exhaustive feature selection compares the performance of all possible feature subsets and chooses the best-performing subset.



Recursive Feature Elimination

Recursive feature elimination starts with the whole feature set and eliminates features repeatedly depending on their relevance as judged by the learning algorithm.

Feature Selection



Identify the most relevant features and discard the rest to reduce noise and overfitting.

Embedded Approach



Regularization

Regularization is a method that adds a penalty term to the loss function to prevent overfitting in machine learning models.



Random Forest Importance

Random Forest computes a feature significance score for each feature as part of the tree-building process, which may be used to order features based on their relevance.

Unsupervised Feature Selection Techniques



Principal Component Analysis (PCA)

PCA is a linear dimensionality reduction method that converts the original feature space into a new orthogonal space defined by principal components.



Independent Component Analysis (ICA)

ICA can be used to convert the original feature space into a new space characterized by statistically independent components.



Non-Negative Matrix Factorization (NMF)

The non-negative matrix factor (NMF) approximates a non-negative data matrix as the product of two lower-dimensional non-negative matrices.



t-distributed Stochastic Neighbor Embedding (t-SNE)

t-SNE is a nonlinear dimensionality reduction method that tries to preserve the dataset's structure by reducing the difference between pairwise probability distributions in high and low-dimensional locations.

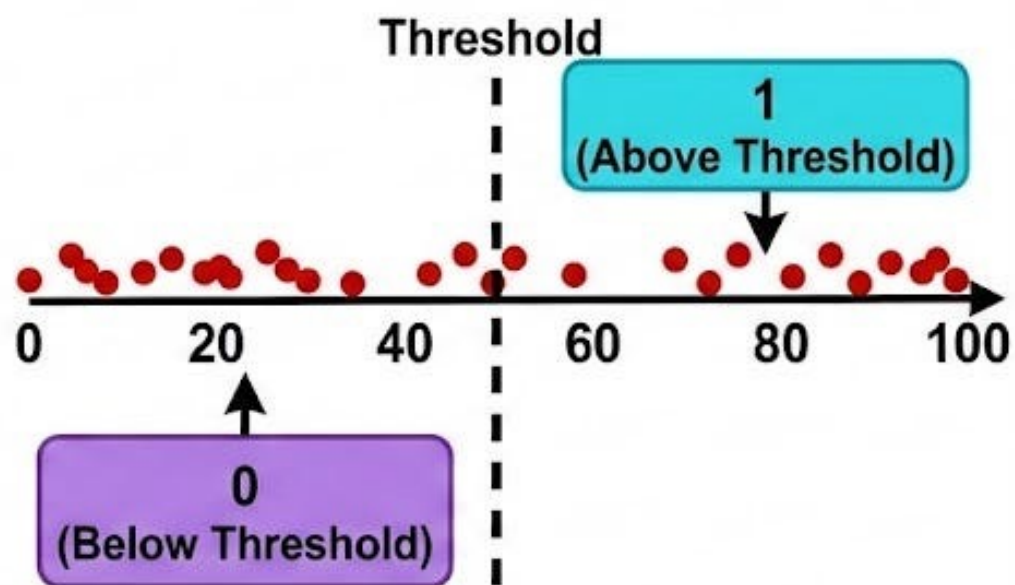


Autoencoder

An autoencoder, which is a kind of artificial neural network, learns to encode input data into a lower-dimensional representation and then decode it back to the original version.

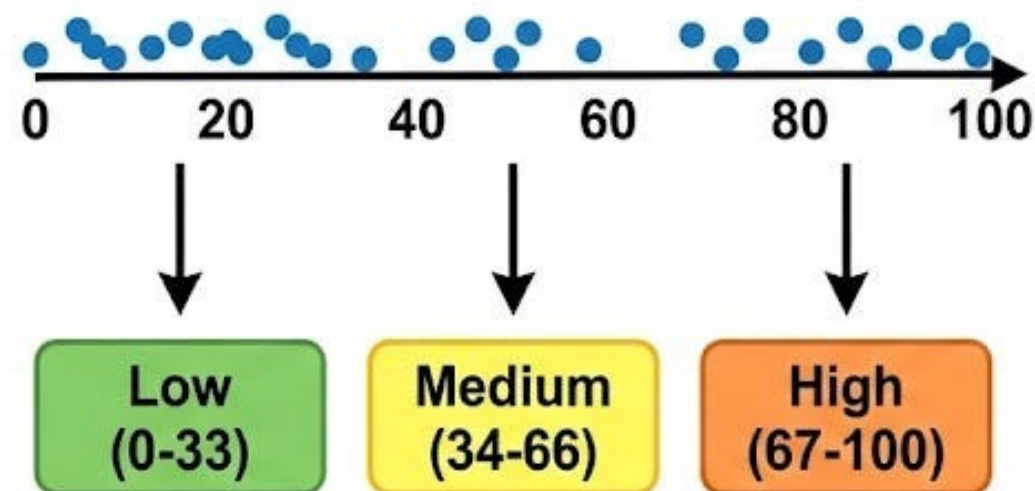
Binary encoding and Discretization ✓

BINARIZATION (Binary Encoding)



Continuous to Binary (0 or 1)
based on a Threshold

BINNING (Discretization)



Continuous to Ordered Categories
(e.g., Equal-Width or Quantile)

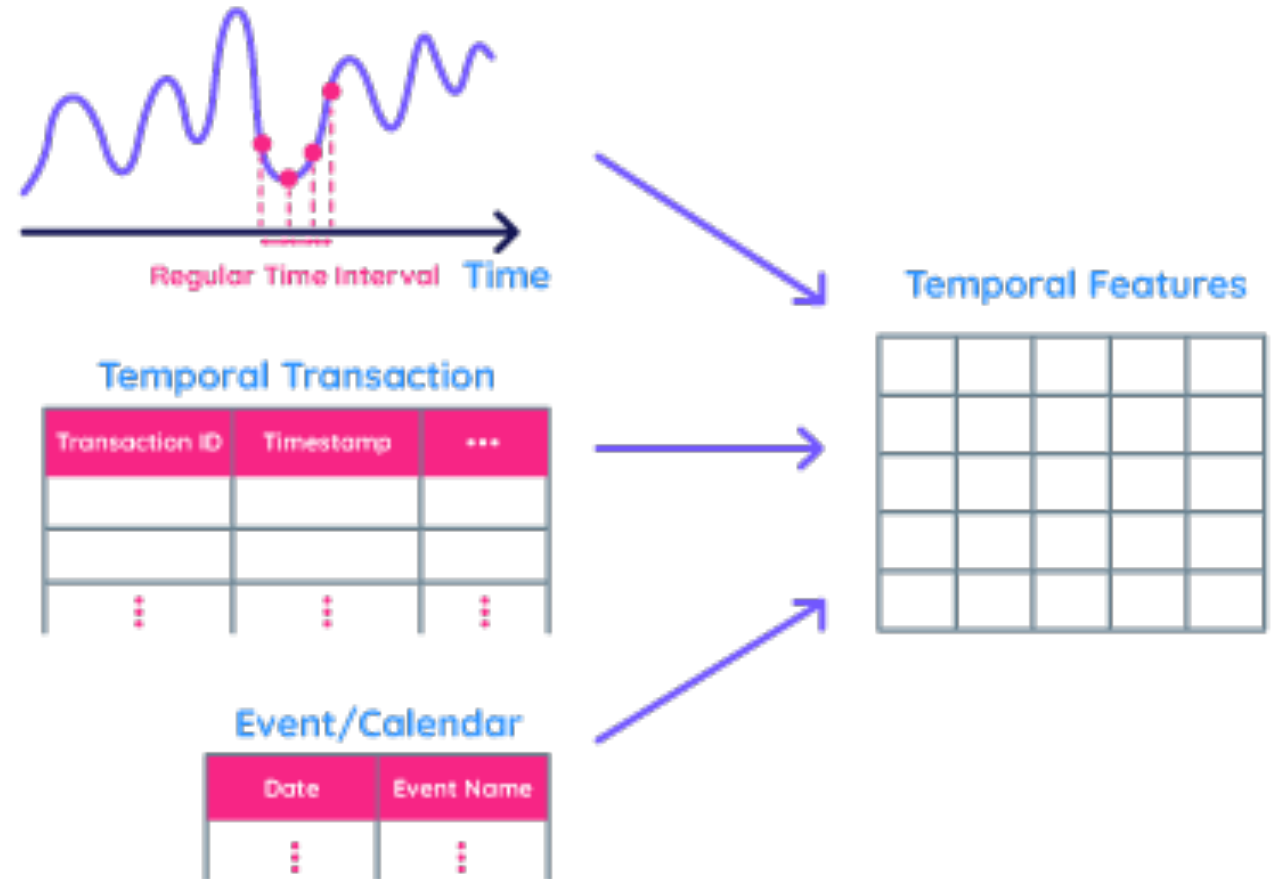
Temporal and Sequential Features



Capture time-dependent patterns by engineering lag variables, rolling averages, and time-since-event features.

Example for sales forecasting:

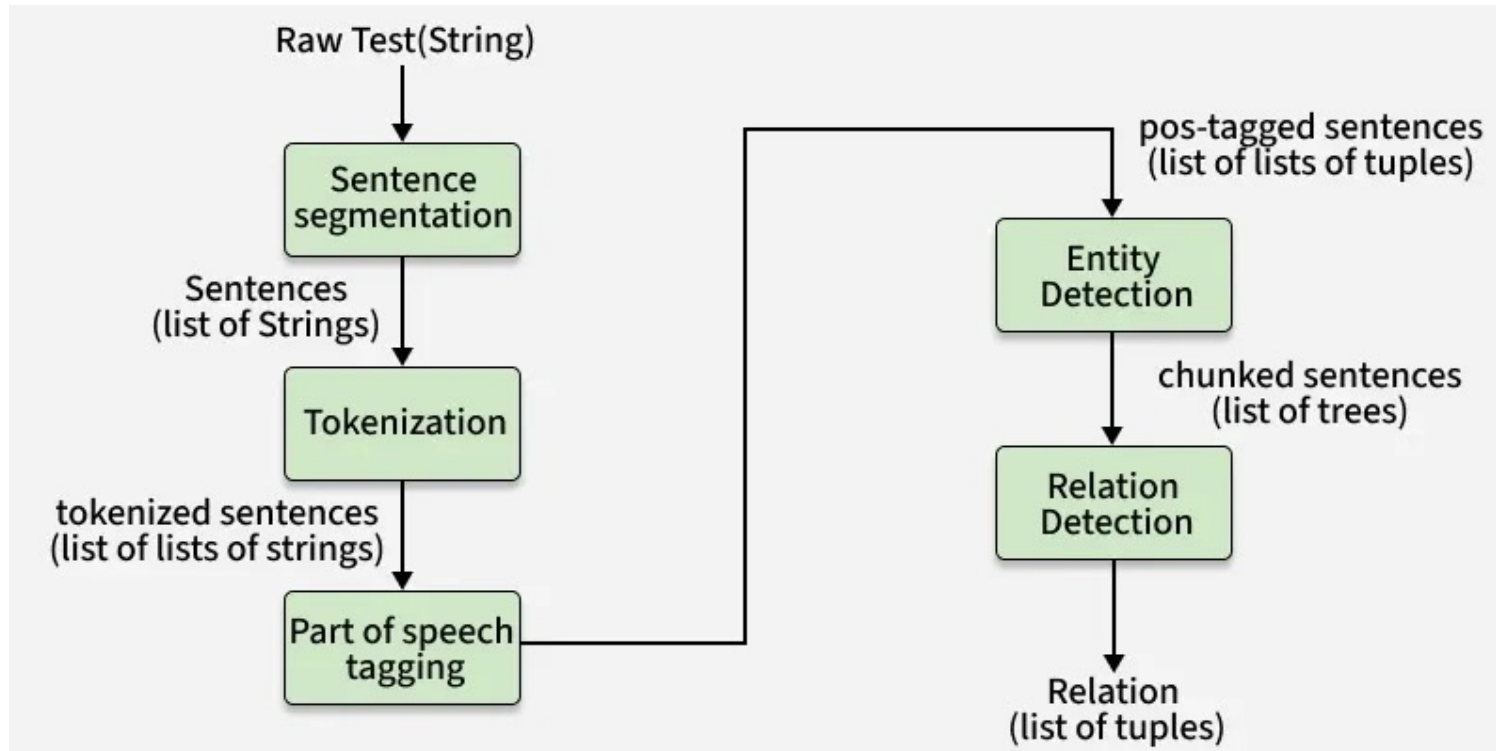
- Create features like sales from 7 days ago.
- 30-day moving average.
- Days since last promotion.



Text and NLP Features



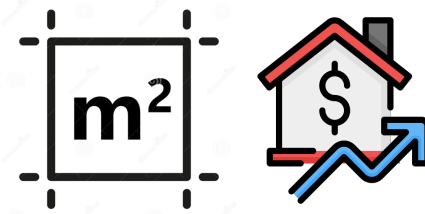
Transform unstructured text into structured numerical representations



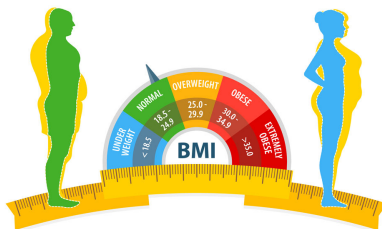
Domain-Specific Transformations



Apply expert knowledge to create features that reflect real-world relationships.



Computing Price per m2



Calculating the body mass index (BMI)
by combining height and weight



RECENCY

The frequency
of the customer
transactions or visits

E.g. Time since last
order or last engaged
with the product



FREQUENCY

The frequency
of the customer
transactions or visits

E.g. Total number of
transactions or
average time between
transactions/ engaged
visits



MONETARY

The intention of
customer to spend or
purchasing power of
customer

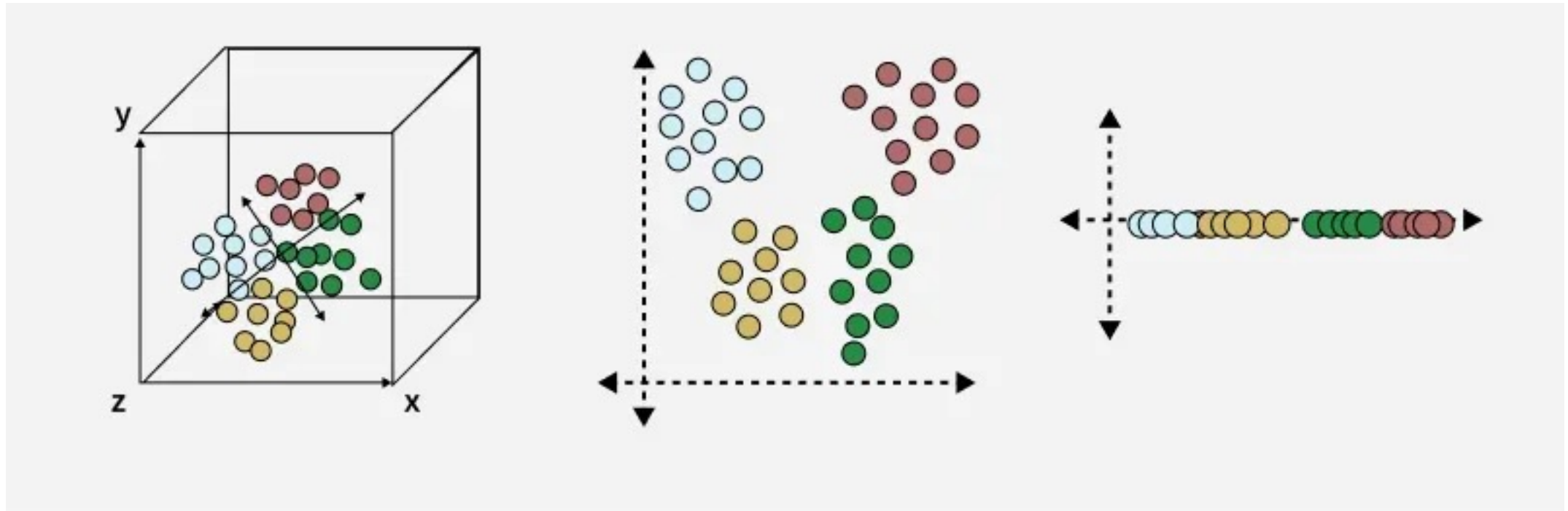
E.g. Total or average
transactions value

Customer segmentation

Dimensionality Reduction



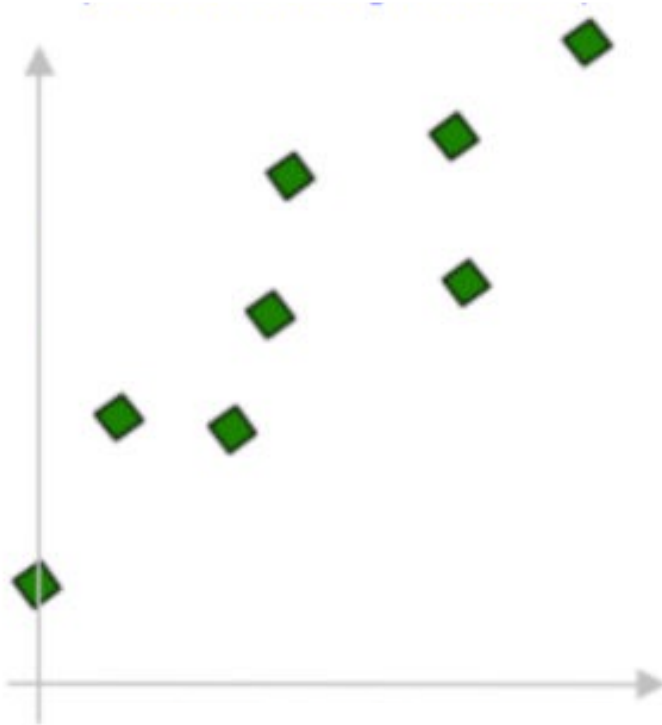
Compress high-dimensional feature spaces into fewer variables while retaining the most important information.



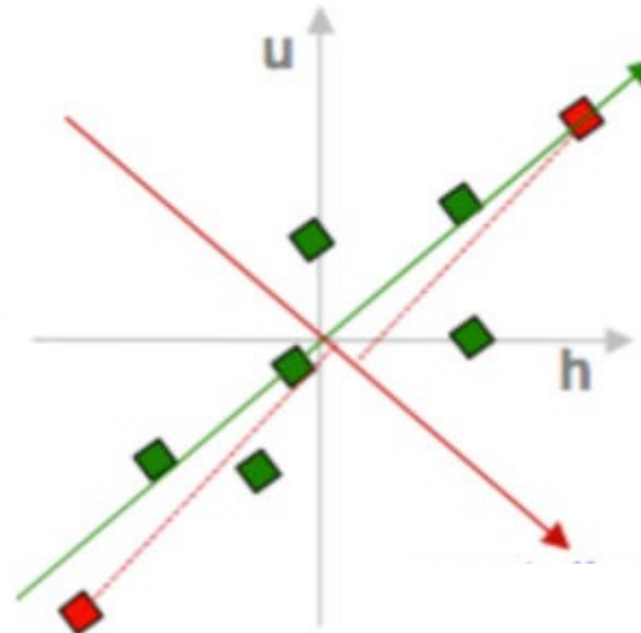
Dimensionality Reduction - PCA



1. correlated hi-d data



2. center the points



3. compute covariance matrix

$$\begin{matrix} & h & u \\ h & \begin{pmatrix} 2.0 & 0.8 \end{pmatrix} \\ u & \begin{pmatrix} 0.8 & 0.6 \end{pmatrix} \end{matrix} \rightarrow \text{cov}(h, u) = \frac{1}{n} \sum_{i=1}^n h_i u_i$$



4. eigenvectors + eigenvalues

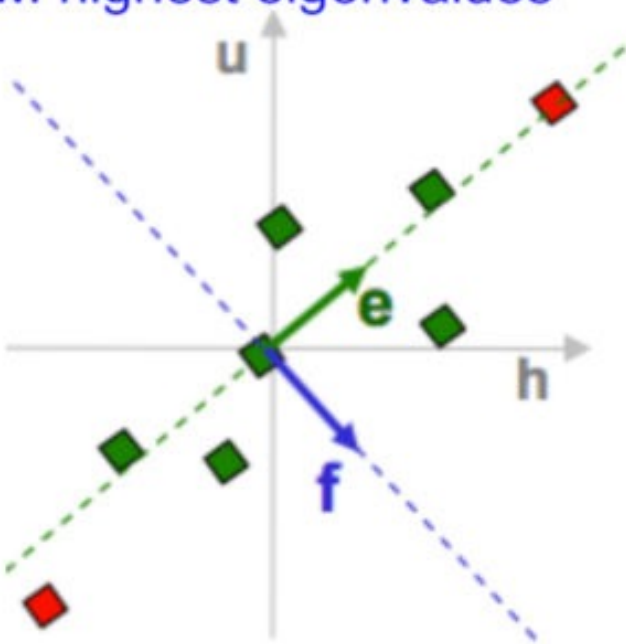
$$\begin{pmatrix} 2.0 & 0.8 \\ 0.8 & 0.6 \end{pmatrix} \begin{bmatrix} e_h \\ e_u \end{bmatrix} = \lambda_e \begin{bmatrix} e_h \\ e_u \end{bmatrix}$$

$$\begin{pmatrix} 2.0 & 0.8 \\ 0.8 & 0.6 \end{pmatrix} \begin{bmatrix} f_h \\ f_u \end{bmatrix} = \lambda_f \begin{bmatrix} f_h \\ f_u \end{bmatrix}$$

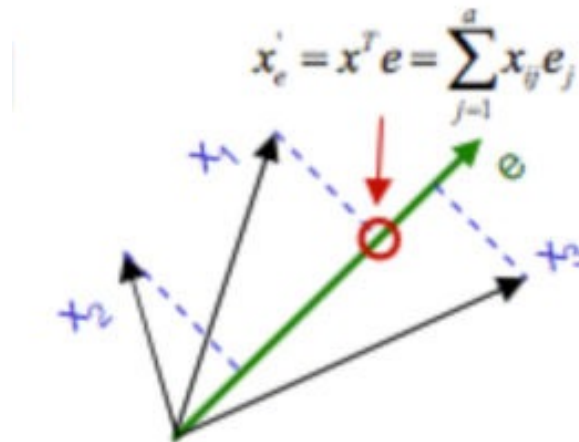
Dimensionality Reduction - PCA



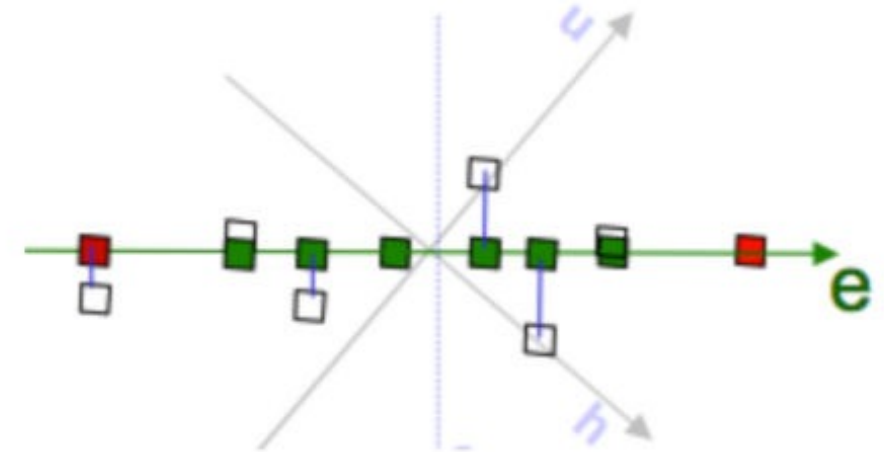
5. pick $m < d$ eigenvectors
w. highest eigenvalues



6. project data points to
those eigenvectors



7. uncorrelated low-d data





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